

Patient Portal - Functional Requirements Document

1. Document Overview

1.1 Purpose

This document defines the functional requirements for the Patient Portal within the Sarthi Platform, enabling patients to manage their healthcare needs, access medical information, communicate with providers, and engage with their care team.

1.2 Scope

The Patient Portal provides patients with comprehensive access to their health information, appointment management, billing, lab results, secure messaging, and AI-powered communication assistance.

1.3 Stakeholders

- **Primary Users:** Patients and authorized caregivers
 - **Secondary Users:** Healthcare providers (for patient communication), Administrative staff (for support)
 - **System Integrators:** EMR vendors, Payment processors, Laboratory systems
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2. Functional Requirements

FR-PP-001: Patient Dashboard and Navigation

Requirement: The system SHALL provide a personalized patient dashboard with navigation to all portal features and health information summary.

User Story: As a patient, I want to see a summary of my health information and quickly access different portal features so that I can manage my healthcare efficiently.

Acceptance Criteria:

- AC-PP-001.1: Dashboard MUST display personalized welcome message with patient name
- AC-PP-001.2: Dashboard MUST show upcoming appointments count and next appointment date
- AC-PP-001.3: Dashboard MUST display unread messages count and sender information
- AC-PP-001.4: Dashboard MUST show outstanding bills total amount and count
- AC-PP-001.5: Dashboard MUST provide navigation sidebar with all portal sections
- AC-PP-001.6: Dashboard MUST highlight active section in navigation

- AC-PP-001.7: Dashboard MUST be responsive and usable on mobile devices (320px minimum width)
- AC-PP-001.8: Dashboard MUST load within 800ms on standard internet connections
- AC-PP-001.9: Dashboard MUST update summary cards when underlying data changes

Priority: High

FR-PP-002: At-Home Device Management

Requirement: The system SHALL display assigned at-home monitoring devices with return dates and usage instructions.

User Story: As a patient, I want to see which medical devices I have checked out and when they're due back so that I can manage my monitoring responsibilities properly.

Acceptance Criteria:

- AC-PP-002.1: Dashboard MUST display "My At-Home Devices" section only when devices are assigned
- AC-PP-002.2: System MUST show device name, type, and assigned date
- AC-PP-002.3: System MUST display return due date with human-friendly relative time
- AC-PP-002.4: System MUST highlight overdue devices with red styling and urgent messaging
- AC-PP-002.5: System MUST provide clear usage instructions for each device type
- AC-PP-002.6: System MUST calculate days until due date accurately across time zones
- AC-PP-002.7: System MUST show devices due within 3 days with yellow warning styling
- AC-PP-002.8: System MUST hide device section when no devices are assigned
- AC-PP-002.9: System MUST sync device assignments in real-time with inventory system

Priority: Medium

FR-PP-003: Quick Actions Interface

Requirement: The system SHALL provide quick access buttons for common patient tasks and navigation to key portal features.

User Story: As a patient, I want quick buttons for the most common tasks so that I can accomplish what I need without navigating through multiple pages.

Acceptance Criteria:

- AC-PP-003.1: System MUST provide "Book a New Appointment" quick action
- AC-PP-003.2: System MUST provide "View Lab Results" quick action
- AC-PP-003.3: System MUST provide "Update Your Profile" quick action
- AC-PP-003.4: System MUST provide "Contact the Office" quick action
- AC-PP-003.5: Quick actions MUST be arranged in 2x2 grid on desktop
- AC-PP-003.6: Quick actions MUST stack vertically on mobile devices
- AC-PP-003.7: All quick action buttons MUST be keyboard accessible
- AC-PP-003.8: Quick actions MUST have appropriate hover states and visual feedback
- AC-PP-003.9: Quick actions MUST navigate to correct portal sections or trigger appropriate workflows

Priority: Medium

FR-PP-004: Appointment Management

Requirement: The system SHALL allow patients to view, book, reschedule, and cancel appointments with real-time availability checking.

User Story: As a patient, I want to manage my appointments online so that I can schedule care at my convenience without calling the office.

Acceptance Criteria:

- AC-PP-004.1: System MUST display all upcoming appointments with date, time, provider, and location
- AC-PP-004.2: System MUST show appointment status (confirmed, needs confirmation, cancelled)
- AC-PP-004.3: System MUST provide "I'm Here" check-in button within 30-minute window before appointment
- AC-PP-004.4: System MUST allow appointment rescheduling when permitted by practice policy
- AC-PP-004.5: System MUST allow appointment cancellation with appropriate notice requirements
- AC-PP-004.6: System MUST provide appointment booking wizard with provider selection
- AC-PP-004.7: System MUST show real-time availability and prevent double-booking
- AC-PP-004.8: System MUST verify insurance coverage for selected appointments
- AC-PP-004.9: System MUST send confirmation notifications via patient's preferred method
- AC-PP-004.10: System MUST display appointment history with visit summaries when available

Priority: High

FR-PP-005: Digital Check-In Process

Requirement: The system SHALL provide digital check-in capability for patients arriving for appointments with insurance verification and form completion.

User Story: As a patient, I want to check in for my appointment using my phone so that I can reduce waiting time and minimize contact at the front desk.

Acceptance Criteria:

- AC-PP-005.1: System MUST provide "I'm Here" button 30 minutes before appointment time
- AC-PP-005.2: System MUST verify patient identity before allowing check-in
- AC-PP-005.3: System MUST confirm current insurance information during check-in
- AC-PP-005.4: System MUST prompt completion of required forms and questionnaires
- AC-PP-005.5: System MUST update patient status in clinic workflow systems
- AC-PP-005.6: System MUST provide estimated wait time after check-in
- AC-PP-005.7: System MUST disable check-in button outside of allowed time window
- AC-PP-005.8: System MUST handle check-in for patients without smartphones via kiosk mode
- AC-PP-005.9: System MUST notify clinic staff immediately upon patient check-in

Priority: High

FR-PP-006: Billing and Payment Management

Requirement: The system SHALL display outstanding bills and payment history with secure online payment processing capabilities.

User Story: As a patient, I want to view and pay my medical bills online so that I can manage my healthcare expenses conveniently and securely.

Acceptance Criteria:

- AC-PP-006.1: System MUST display outstanding bills with service date, description, and amount due
- AC-PP-006.2: System MUST highlight overdue bills with appropriate visual indicators
- AC-PP-006.3: System MUST show payment history with payment dates, amounts, and methods
- AC-PP-006.4: System MUST provide secure payment processing for credit cards and bank accounts

- AC-PP-006.5: System MUST support partial payments when enabled by practice
- AC-PP-006.6: System MUST generate payment confirmations and receipts
- AC-PP-006.7: System MUST allow payment method saving for future use (with patient consent)
- AC-PP-006.8: System MUST integrate with practice billing systems for real-time updates
- AC-PP-006.9: System MUST provide downloadable receipt and statement options
- AC-PP-006.10: System MUST comply with PCI DSS requirements for payment processing

Priority: High

FR-PP-007: Lab Results Access

Requirement: The system SHALL display lab results with appropriate clinical safeguards and provider context when available.

User Story: As a patient, I want to access my lab results as soon as they're available so that I can stay informed about my health status.

Acceptance Criteria:

- AC-PP-007.1: System MUST display only lab results approved for patient release by physicians
- AC-PP-007.2: System MUST show normal lab results with "View Details" option
- AC-PP-007.3: System MUST show abnormal lab results with "Contact Office" option instead of details
- AC-PP-007.4: System MUST provide detailed lab values with reference ranges for normal results
- AC-PP-007.5: System MUST include provider interpretation and notes when available
- AC-PP-007.6: System MUST organize results by test date with most recent first
- AC-PP-007.7: System MUST provide filtering by test type and date range
- AC-PP-007.8: System MUST allow downloading or printing of lab results
- AC-PP-007.9: System MUST prevent access to results not yet reviewed by provider

Priority: High

FR-PP-008: Secure Messaging System

Requirement: The system SHALL provide HIPAA-compliant secure messaging between patients and healthcare providers with file attachments and message threading.

User Story: As a patient, I want to send secure messages to my healthcare team so that I can ask questions and receive guidance without scheduling appointments for routine concerns.

Acceptance Criteria:

- AC-PP-008.1: System MUST provide secure, encrypted messaging interface
- AC-PP-008.2: System MUST organize messages in threaded conversations by topic
- AC-PP-008.3: System MUST allow file attachments (PDFs and images only, max 10MB)
- AC-PP-008.4: System MUST provide message categories (general, appointment, billing, prescription)
- AC-PP-008.5: System MUST show message read status and provider response times
- AC-PP-008.6: System MUST provide rich text formatting for message composition
- AC-PP-008.7: System MUST route messages to appropriate care team members
- AC-PP-008.8: System MUST provide urgent message flagging for time-sensitive concerns
- AC-PP-008.9: System MUST maintain complete message audit trail for compliance
- AC-PP-008.10: System MUST automatically expire messages after configurable time period

Priority: High

FR-PP-009: AI Communication Assistant

Requirement: The system SHALL provide 24/7 AI-powered chat assistance for administrative questions with appropriate escalation to human staff and strict medical advice boundaries.

User Story: As a patient, I want to get quick answers to basic questions any time of day so that I don't have to wait for office hours for routine administrative matters.

Acceptance Criteria:

- AC-PP-009.1: System MUST provide floating chat button accessible from all portal pages
- AC-PP-009.2: System MUST clearly identify conversations as AI-assisted in chat interface
- AC-PP-009.3: System MUST handle administrative questions (appointments, billing, office hours, directions)
- AC-PP-009.4: System MUST refuse to provide medical advice and redirect to appropriate channels
- AC-PP-009.5: System MUST detect emergency keywords and provide immediate 911/emergency guidance
- AC-PP-009.6: System MUST escalate to human staff during office hours when requested
- AC-PP-009.7: System MUST take detailed messages for staff follow-up during after-hours

- AC-PP-009.8: System MUST provide different service levels for office hours vs after-hours
- AC-PP-009.9: System MUST maintain conversation logs for quality assurance and training
- AC-PP-009.10: System MUST comply with HIPAA requirements for patient communication

Priority: Medium

FR-PP-010: AI Chat Office Hours vs After-Hours Behavior

Requirement: The system SHALL provide different AI assistance capabilities during office hours versus after-hours with appropriate escalation protocols.

User Story: As a patient, I want to know what level of assistance is available when I use the chat feature so that I have appropriate expectations for response and escalation.

Acceptance Criteria:

- AC-PP-010.1: System MUST offer human transfer within 5 minutes during office hours
- AC-PP-010.2: System MUST provide extended AI assistance (up to 15 minutes) during after-hours
- AC-PP-010.3: System MUST clearly communicate office status in chat interface
- AC-PP-010.4: System MUST collect detailed messages for next business day follow-up after hours
- AC-PP-010.5: System MUST detect and escalate emergency situations immediately regardless of time
- AC-PP-010.6: System MUST provide emergency contact numbers for after-hours emergencies
- AC-PP-010.7: System MUST offer to schedule callbacks for complex issues during after-hours
- AC-PP-010.8: System MUST notify staff of urgent after-hours messages via configured channels
- AC-PP-010.9: System MUST maintain different response time expectations for office vs after-hours

Priority: Medium

FR-PP-011: Profile Management

Requirement: The system SHALL allow patients to view and update personal information, preferences, and medical history with appropriate validation and verification.

User Story: As a patient, I want to keep my contact information and preferences current so that the healthcare team can reach me and provide personalized care.

Acceptance Criteria:

- AC-PP-011.1: System MUST display current personal information (name, address, phone, email)

- AC-PP-011.2: System MUST allow editing of contact information with validation
- AC-PP-011.3: System MUST require identity verification for sensitive changes (DOB, SSN)
- AC-PP-011.4: System MUST provide communication preference settings (email, SMS, phone)
- AC-PP-011.5: System MUST allow patients to update emergency contact information
- AC-PP-011.6: System MUST provide insurance information management
- AC-PP-011.7: System MUST allow patients to report medication changes and allergies
- AC-PP-011.8: System MUST maintain audit trail of all profile changes
- AC-PP-011.9: System MUST require two-factor authentication for critical information changes

Priority: Medium

FR-PP-012: Mobile Responsiveness and Accessibility

Requirement: The system SHALL provide full functionality on mobile devices with accessibility compliance for users with disabilities.

User Story: As a patient, I want to use the portal on my smartphone with the same functionality as on desktop so that I can manage my healthcare from anywhere.

Acceptance Criteria:

- AC-PP-012.1: System MUST be fully functional on smartphones (320px minimum width)
- AC-PP-012.2: System MUST provide touch-friendly interface elements (44px minimum touch targets)
- AC-PP-012.3: System MUST adapt navigation for mobile with collapsible menu
- AC-PP-012.4: System MUST optimize loading times for mobile networks
- AC-PP-012.5: System MUST comply with WCAG 2.1 AA accessibility standards
- AC-PP-012.6: System MUST provide keyboard navigation for all features
- AC-PP-012.7: System MUST support screen readers with appropriate ARIA labels
- AC-PP-012.8: System MUST provide high contrast mode for visually impaired users
- AC-PP-012.9: System MUST work offline for critical functions with data sync when connected

Priority: High

FR-PP-013: Real-Time Data Synchronization

Requirement: The system SHALL synchronize patient data in real-time across all systems and provide immediate updates when changes occur.

User Story: As a patient, I want to see updated information immediately when my healthcare team makes changes so that I always have the most current information.

Acceptance Criteria:

- AC-PP-013.1: System MUST update appointment status within 30 seconds of changes
- AC-PP-013.2: System MUST display new lab results within 5 minutes of provider approval
- AC-PP-013.3: System MUST show new messages within 1 minute of provider sending
- AC-PP-013.4: System MUST update billing information within 15 minutes of payment posting
- AC-PP-013.5: System MUST sync device assignments immediately when assigned by staff
- AC-PP-013.6: System MUST handle concurrent access across multiple browser tabs
- AC-PP-013.7: System MUST resolve data conflicts with server data taking precedence
- AC-PP-013.8: System MUST provide visual indicators for pending data synchronization
- AC-PP-013.9: System MUST queue updates during offline periods and sync when reconnected

Priority: High

FR-PP-014: Multi-Language Support

Requirement: The system SHALL support multiple languages with culturally appropriate content and right-to-left text rendering where applicable.

User Story: As a patient who speaks a language other than English, I want to use the portal in my preferred language so that I can understand my healthcare information clearly.

Acceptance Criteria:

- AC-PP-014.1: System MUST support English, Spanish, Chinese, Vietnamese, Arabic, and Russian
- AC-PP-014.2: System MUST provide complete translation of all interface elements
- AC-PP-014.3: System MUST translate AI chat responses into patient's selected language
- AC-PP-014.4: System MUST support right-to-left text rendering for Arabic
- AC-PP-014.5: System MUST translate care notes and health education materials
- AC-PP-014.6: System MUST remember language preference across sessions
- AC-PP-014.7: System MUST provide cultural adaptations for date/time and number formats

- AC-PP-014.8: System MUST offer professional medical translation for complex content
- AC-PP-014.9: System MUST allow language switching without losing current page context

Priority: Medium

3. Non-Functional Requirements

NFR-PP-001: Security and Privacy

- System MUST encrypt all patient data in transit using TLS 1.3 minimum
- System MUST encrypt all stored PHI using AES-256 encryption
- System MUST implement session timeouts of 30 minutes for inactive sessions
- System MUST log all patient data access for HIPAA compliance audit trails
- System MUST implement multi-factor authentication for sensitive account changes
- System MUST comply with HIPAA Privacy and Security Rules
- System MUST provide patient consent management for data sharing

NFR-PP-002: Performance and Reliability

- System MUST load initial dashboard within 800ms on broadband connections
- System MUST handle 10,000 concurrent patient sessions across all practices
- System MUST maintain 99.9% uptime during peak usage hours (8 AM - 6 PM)
- System MUST provide graceful degradation when backend services are unavailable
- System MUST cache frequently accessed data for improved performance
- System MUST implement progressive loading for large data sets
- System MUST support offline functionality for critical features

NFR-PP-003: Usability and User Experience

- System MUST be usable by patients with basic computer literacy
- System MUST provide consistent navigation patterns across all portal sections
- System MUST offer contextual help and tooltips for complex features
- System MUST support undo functionality for reversible actions
- System MUST provide clear error messages with suggested resolution steps
- System MUST implement responsive design for seamless cross-device experience
- System MUST maintain user interface consistency with healthcare industry standards

NFR-PP-004: Integration and Interoperability

- System MUST integrate with EMR systems via standardized APIs (FHIR preferred)
 - System MUST support real-time data exchange with practice management systems
 - System MUST integrate with payment processors meeting PCI DSS standards
 - System MUST support single sign-on (SSO) integration where required
 - System MUST provide API endpoints for third-party integrations
 - System MUST support data export in standard formats (PDF, CSV, HL7)
 - System MUST maintain data consistency across integrated systems
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4. Constraints and Assumptions

Technical Constraints

- Must support modern web browsers (Chrome, Firefox, Safari, Edge) - latest 2 versions
- Must work on mobile devices with iOS 14+ and Android 10+
- Must integrate with existing practice EMR and billing systems
- Must comply with healthcare data exchange standards (FHIR, HL7)
- Must use cloud infrastructure with healthcare compliance certifications

Business Constraints

- Must not duplicate functionality already available in EMR systems
- Must provide immediate value to patients to encourage adoption
- Must reduce administrative burden on healthcare staff
- Must support multiple practice configurations and workflows
- Must scale to support large healthcare organizations

Regulatory Constraints

- Must comply with HIPAA Privacy and Security Rules
- Must meet state and federal healthcare regulations
- Must support patient rights under healthcare privacy laws
- Must provide audit capabilities for regulatory compliance
- Must support data retention policies as required by law

User Constraints

- Must accommodate patients with varying levels of technical literacy
 - Must support patients with disabilities through accessibility features
 - Must work for patients with limited internet connectivity
 - Must support patients who prefer different languages
 - Must accommodate patients who use assistive technologies
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5. Acceptance Criteria Summary

The Patient Portal system will be considered complete when:

1. All functional requirements (FR-PP-001 through FR-PP-014) are implemented and tested
2. All acceptance criteria are met and verified through user acceptance testing with actual patients
3. Security and HIPAA compliance requirements are validated by third-party security audit
4. Performance benchmarks are achieved under simulated patient load testing
5. Integration testing with EMR, billing, and laboratory systems is successful
6. Accessibility compliance is verified through independent accessibility testing
7. Multi-language functionality is tested and validated by native speakers
8. Mobile responsiveness is confirmed across target devices and browsers
9. AI chat functionality is tested for appropriate medical boundary enforcement
10. Patient feedback from beta testing indicates satisfactory user experience and adoption readiness